

SKILL 9

Question 9.1 — Change Directories

Question:

To change directories, validate paths, update the working directory, handle errors, maintain the previous directory, and test navigation.

Question 9.2 — Built-in Command Dispatch

Question:

To identify built-ins, create a dispatch table, execute in-process commands, handle invalid commands, maintain state, and test dispatch logic.

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano skill9.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc skill9.c -o skill9
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./skill9
Current directory: /home/saikrishna__90156
Enter directory to change to: Krishna_144
Directory change failed: No such file or directory
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./skill9
Current directory: /home/saikrishna__90156
Enter directory to change to: /tmp
Directory changed successfully.
New directory: /tmp
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano skill9a.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc skill9a.c -o skill9a
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./skill9a
shell> ls
Unknown command: ls
shell> help
Built-in commands:
help
pwd
exit
shell> pwd
/home/saikrishna__90156
shell> hello
Unknown command: hello
shell> exit
Exiting shell...
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat skill9.c
#include <stdio.h>
#include <unistd.h>
#include <limits.h>
#include <string.h>

int main()
{
```

```

{
    char path[PATH_MAX];

    if (getcwd(path, sizeof(path)) != NULL)
        printf("Current directory: %s\n", path);
    else
        perror("getcwd");

    printf("Enter directory to change to: ");

    if (fgets(path, sizeof(path), stdin) == NULL)
        return 1;

    path[strcspn(path, "\n")] = '\0';

    if (chdir(path) == 0)
    {
        printf("Directory changed successfully.\n");

        if (getcwd(path, sizeof(path)) != NULL)
            printf("New directory: %s\n", path);
    }
    else
    {
        perror("Directory change failed");
    }

    return 0;
}
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat skill9a.c
#include <stdio.h>
#include <string.h>
#include <unistd.h>

void cmd_help()

```

```
        return 0;
    }
    saikrishna__90156@DESKTOP-2JF9H5T:~$ cat skill9a.c
#include <stdio.h>
#include <string.h>
#include <unistd.h>

void cmd_help()
{
    printf("Built-in commands:\n");
    printf("help\n");
    printf("pwd\n");
    printf("exit\n");
}

void cmd_pwd()
{
    char path[1024];

    if (getcwd(path, sizeof(path)) != NULL)
        printf("%s\n", path);
}

int main()
{
    char command[100];

    while (1)
    {
        printf("shell> ");
```



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```
{
    char command[100];

    while (1)
    {
        printf("shell> ");

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "help") == 0)
        {
            cmd_help();
        }
        else if (strcmp(command, "pwd") == 0)
        {
            cmd_pwd();
        }
        else if (strcmp(command, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }
        else
        {
            printf("Unknown command: %s\n", command);
        }
    }

    return 0;
}
```

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